



Career-Compass

All-in-one App

A self-developed application that assists students in career guidance

Group:
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Presentation

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i Project Information available in
[GitHub](#).



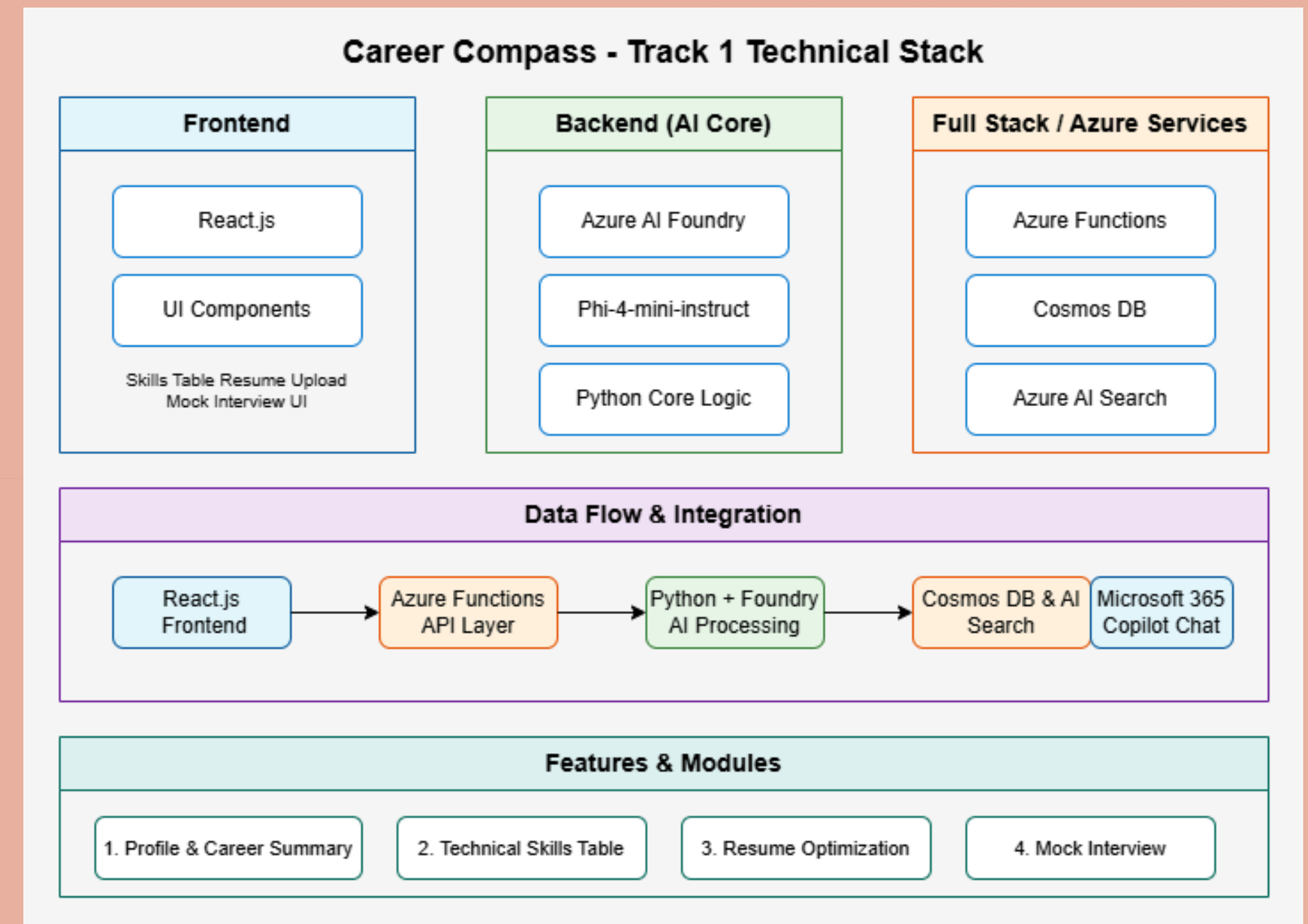
Technical Architecture & Division

1 Frontend

2 Backend

3 Full-Stack

Our implementation drives progress across three distinct technical pillars:



Core features& Functioning

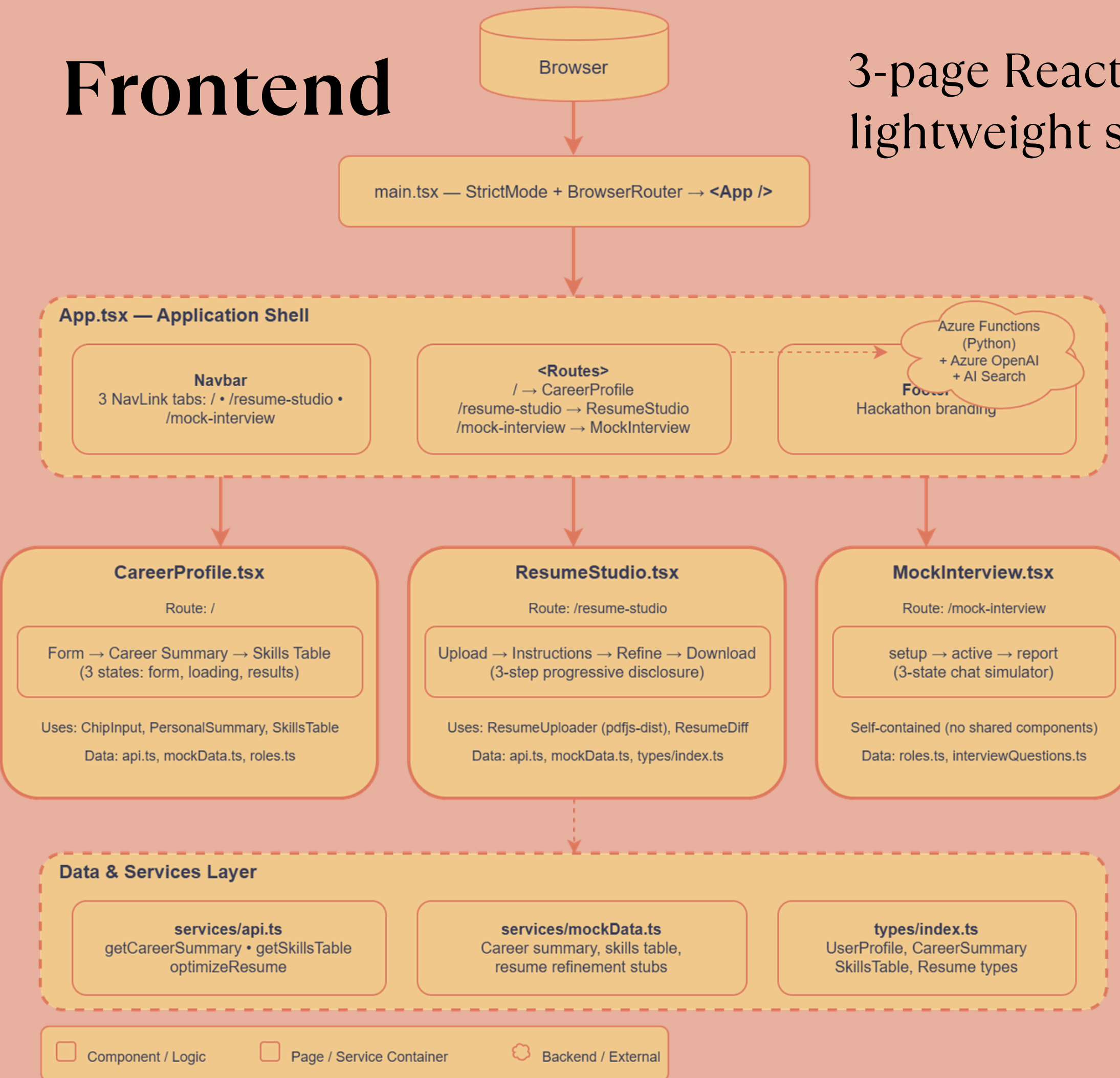
What have we built and what it can do?

- 🎓 Student Profile & Career Summary
- 📊 Technical Skills Table
- 📄 AI Resume Tailoring & Optimization
- 🎤 Interview Question Generator

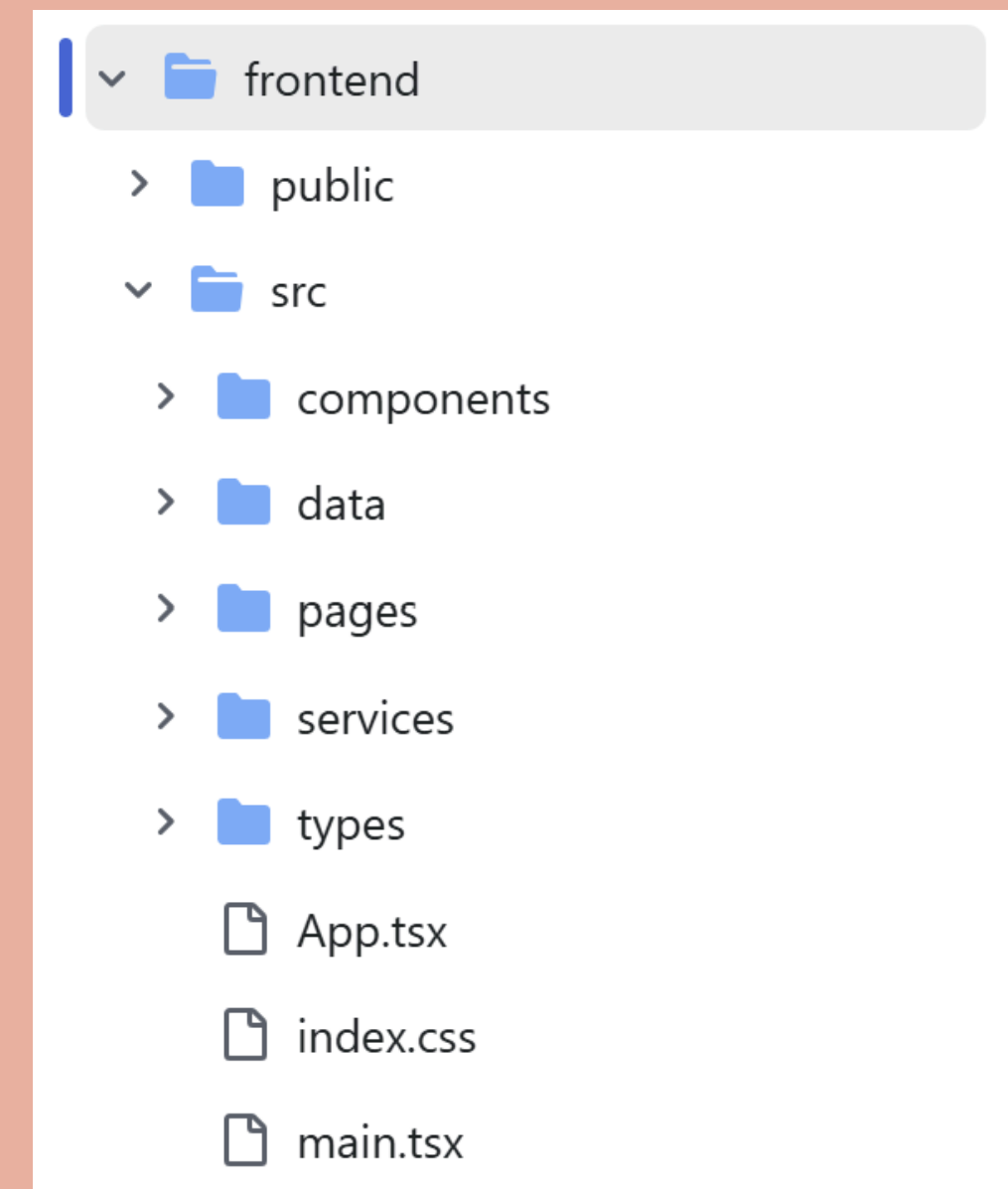
Frontend

3-page React SPA with a Vite + TypeScript lightweight stack and single CSS file

4



- Career Profile: fill a form, get AI-generated career summary + color-coded skills table
- Resume Studio: upload a resume, tell the AI what to improve, see a side-by-side diff
- Mock Interview: chat-based practice with role-specific questions and a scored report

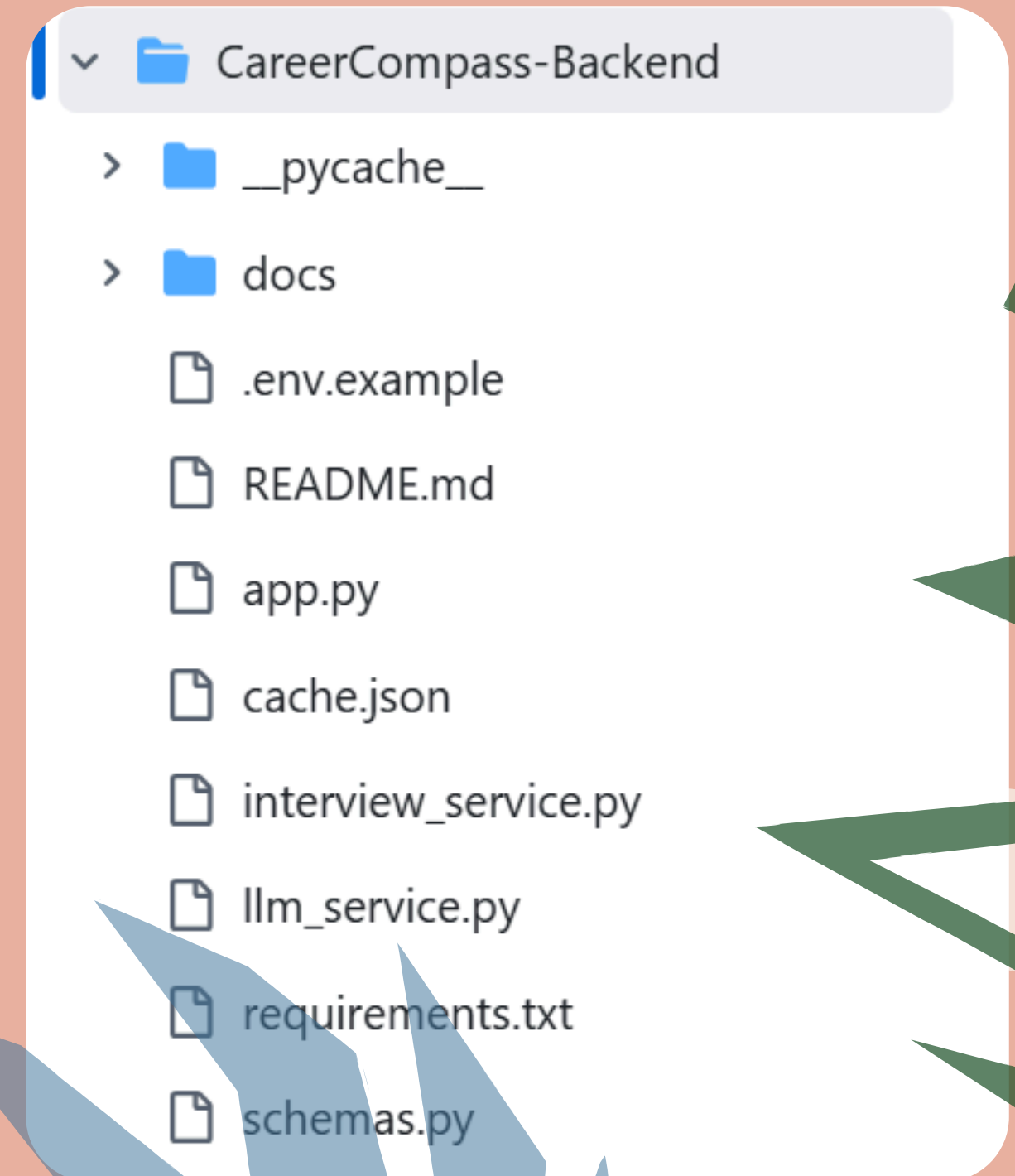


Let's start from the First Part...

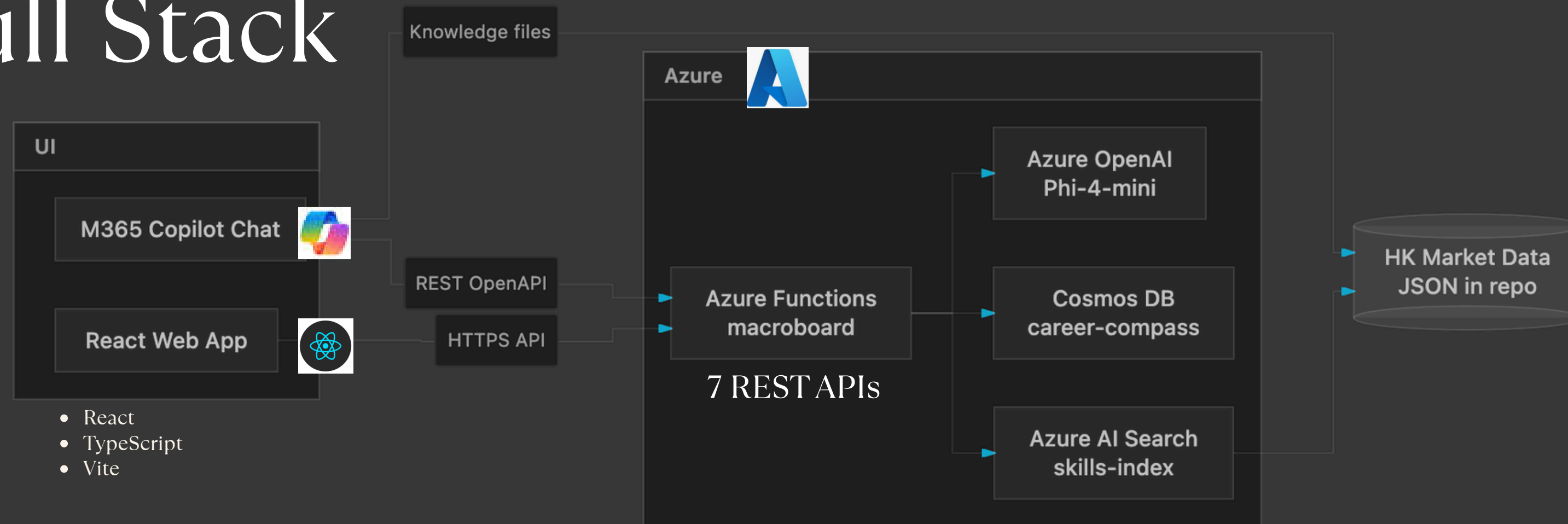
Backend/ AI Core

Architected the core AI orchestration and Azure infrastructure to drive personalized career readiness features.

- Infrastructure & Model Deployment: Deployed Phi-4-mini-instruct on Azure OpenAI to make the three functions achievable.
- Custom Agent & FastAPI Integration: Developed the core AI Agent logic entirely in Python and successfully compiled and executed the pipeline via FastAPI
- Strict JSON Schema Guardrails: Ruled precise prompt logic paired with Strict JSON Outputs to ensure efficient frontend-backend synchronization.



Full Stack



Data

HK-local career data

+

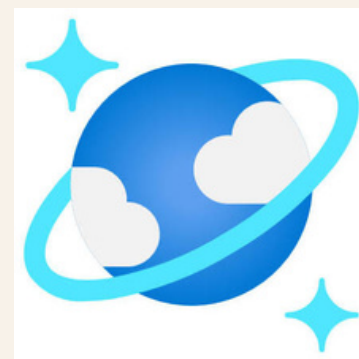
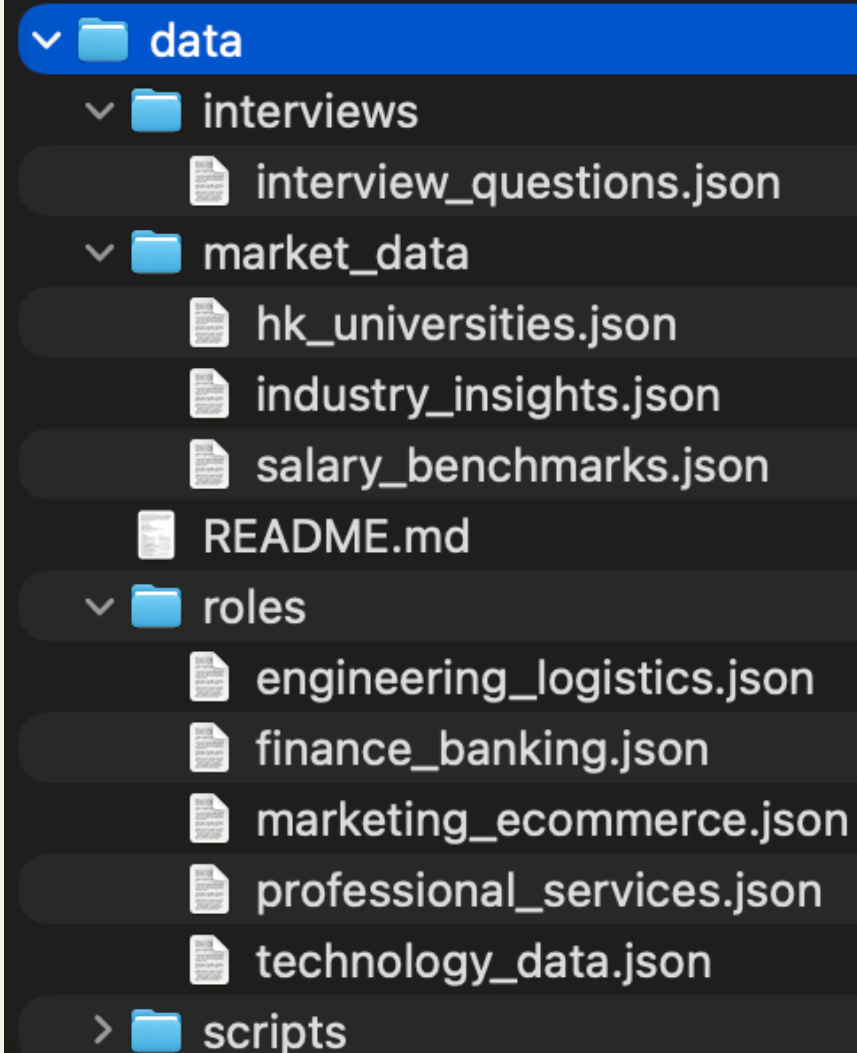
user profiles
persisted in
CosmosDB

+

skills grounded in
structured JSON

+

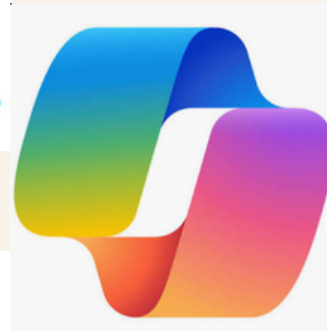
Search.



Knowledge

Provide trusted context to guide decisions

- salary_benchmarks.json ×
- industry_insights.json ×
- skills_summary.json ×



Deployment

- Azure Function App: macroboard
- Live API
- Azure OpenAI: Phi-4-mini-instruct (via hub hub-career-compass-ai)
- Copilot Studio: Career Compass agent + Career Compass API connector (OpenAPI spec in repo)
- CI/CD: func azure functionapp publish macroboard --python + push_app_settings.sh
- Source: GitHub main – backend, data scripts, OpenAPI spec